Borj-PiCare

Ward Control Tower with bedside nodes and a premium Family Care Link

Version 2 | Idea document | Health tech hackathon, Tunisian public hospital wards

> Borj-PiCare helps scarce ward staff go further. It reads vital signs from ward machines, captures bedside requests and nurse confirmations, ranks patients with a deterministic early-warning score, suggests workload rebalancing for a human to approve, and lets the charge nurse test "what if?" scenarios on a digital twin. A consent-based Family Care Link is the premium layer.

Focus	One acute ward (internal medicine or step-down unit)
Core	Ward Control Tower: digital twin, NEWS2 priority, medication timeline, nurse assignment suggestions, what-if simulation
Data capture	Device gateway (USB / Ethernet), low-cost bedside nodes, fast manual entry as fallback
Premium	Family Care Link: consented, nurse-curated updates for relatives
Principle	Advisory only. Deterministic clinical rules. A human approves every reassignment.
Status	Concept stage. Figures are illustrative assumptions, not measured results.

1. Executive summary

Borj-PiCare is a decision-support system for one acute hospital ward. It has three parts:

- **Ward Control Tower (Borj).** A live model of the ward, a deterministic early-warning score (NEWS2), a medication timeline, suggested nurse assignments that a human approves, and a digital twin that answers what-if questions.
- **Data capture.** Vital signs read from ward machines through their USB or Ethernet ports where the data format allows, low-cost bedside nodes for patient requests and nurse confirmations, and a fast entry form for whatever machines cannot provide.
- **Family Care Link (premium).** A consent-based, nurse-curated status window for relatives, and the premium layer of the business model.

Guiding principle: advisory only. Clinical rules are deterministic and explainable, a human approves every reassignment, and the system never chooses drugs or doses.

Scope discipline. One ward, done deeply. Home care, hospital-wide planning and national integration are later phases.

2. The problem

Picture a charge nurse on a night shift in a public internal medicine ward. A few nurses cover many beds with very different levels of clinical acuity. Vitals are on paper charts, medication times are in notebooks, and nothing shows at a glance who is deteriorating. At 3 am two emergency admissions arrive while a patient in bed 6 spikes a fever. She works out in her head who to see first, who covers which room, and which routine task can wait.

Typical pressures (to be validated through ward interviews):

- Few nurses for many patients, with mixed acuity.
- Fragmented or paper-based information, so deterioration is easy to miss between rounds.
- Time lost walking to answer calls that turn out to be non-clinical.
- Disruptions (admissions, a patient worsening, a colleague called away) force constant mental recalculation.
- Phone calls and visits from worried relatives interrupt care.

> Core insight: the budget and the number of nurses cannot double overnight. The practical lever is to make scarce staff go further: fewer wasted trips, earlier warning of deterioration, and fast, explainable rebalancing when the night goes wrong.

3. Solution overview

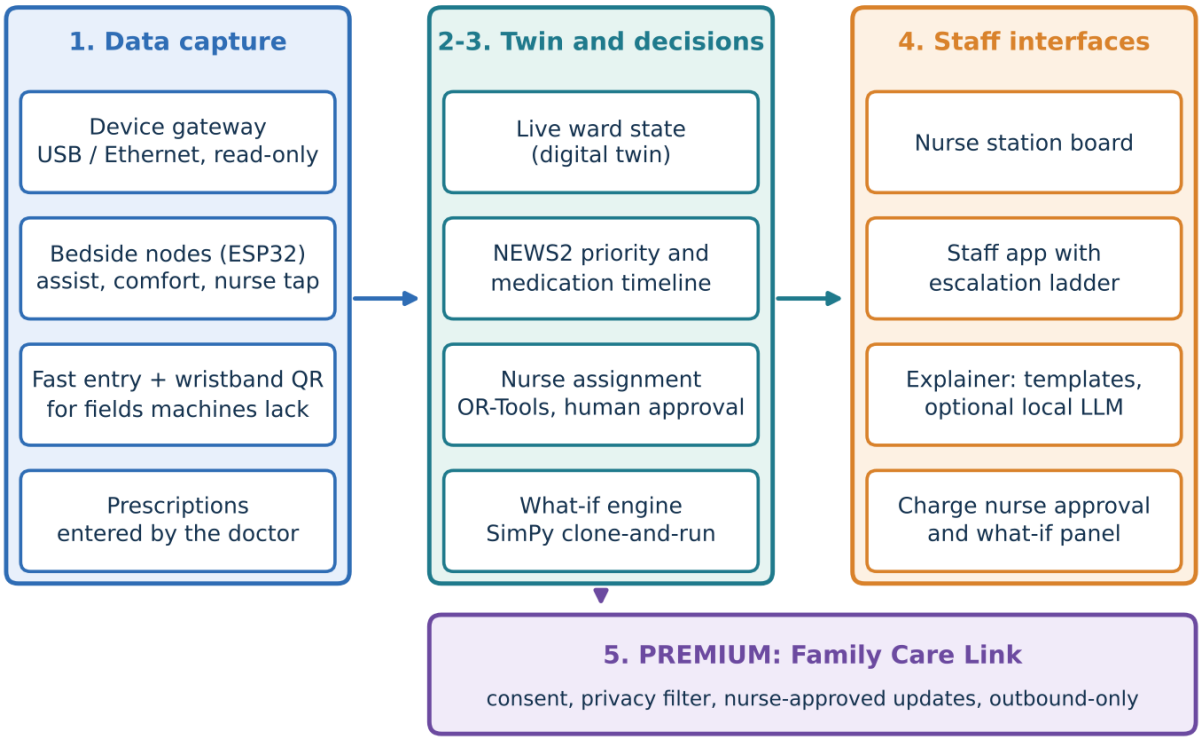


Figure 1. Five layers: data capture feeds the ward twin and decision core, which serve staff interfaces and the premium Family Care Link.

Layer	Contents
1. Data capture	Device gateway, bedside nodes, fast entry with wristband QR, prescription entry
2. Ward twin	Live state model of beds, patients, staff and tasks, plus clone-and-run simulation
3. Decision modules	NEWS2 priority, medication timeline, nurse assignment, equipment reminders
4. Staff interfaces	Station board, staff app with escalation ladder, explainer
5. Premium	Family Care Link

4. Layer 1: data capture

4.1 Vital signs from ward machines

An informal check with a contact reports that many ward machines have USB and Ethernet ports. That makes automatic capture of vital signs realistic, but a port is not the same as an open data format. Each device model must be checked for what the port carries (an open standard such as HL7, a serial data stream, a proprietary protocol, or service-only access), what the vendor and the hospital biomedical team allow, and how the data is tied to a patient.

Design rules

- **Read-only and isolated.** The gateway never writes to a device and sits on its own network segment. The prototype is non-certified and needs hospital biomedical approval before any connection.
- **Common event format.** Every reading becomes an event: device, bed, parameter, value, unit, timestamp, quality flag.
- **Validity and freshness checks.** Sensor-off, artifacts, implausible values and stale readings are flagged, and NEWS2 uses only fresh, valid values.

- **Bed-to-patient binding.** Machines move between patients, so a nurse confirms the association by scanning the wristband at admission and after any transfer or device swap.
- **Fields machines cannot supply.** Consciousness level and supplemental-oxygen status are entered by the nurse, and respiratory rate may need manual entry depending on the monitor. A 10-second form pre-filled with device values covers them.
- **Fallback.** Where a device has no usable port, a 30-second manual form with wristband QR scan takes its place.

Integration survey (to complete on the target ward)

Device type	Typical data	Port and protocol	Access	Priority
Bedside patient monitor	Heart rate, SpO2, blood pressure, respiratory rate, temperature	USB / Ethernet, protocol to verify	Read-only gateway	High
Pulse oximeter	SpO2, pulse	USB or serial, to verify	Read-only gateway	High
Automatic BP monitor	Blood pressure, pulse	To verify	Read-only gateway	Medium
Infusion pump	Status, alarms, rate	Ethernet or USB, to verify	Status only	Later
Thermometer	Temperature	To verify	Read-only gateway	Low

4.2 Bedside node

Each bed gets a low-cost node built on an ESP32: an assist button, a comfort button, a status LED ring, a buzzer, and an NFC reader or QR tag for the nurse tap. An optional small speaker plays pre-recorded phrases. The ESP32 is cheaper and more robust than a Raspberry Pi for this job (low power, fast boot, no SD card to wear out). A Raspberry Pi stays at ward level as gateway and server. Target cost per bed is low, to be confirmed with a real bill of materials.

Call feedback: the patient always knows what happened

State	Patient sees or hears	System behavior
Pressed	Short beep, LED pulses	Event created with a unique ID
Sent	LED steady amber	Delivered to the ward gateway and the nurse station board
Acknowledged	LED pulses green	A nurse accepted the call and someone is coming
Not sent (no link within a few seconds)	Fast red blink, local buzzer, recorded phrase telling the patient to use the emergency bell or call out	Event stored locally and sent when the link returns; ward beacon alerts if installed
Unacknowledged too long	LED stays amber	Re-alert, then escalate to the charge nurse after a set time

- **Never the only route.** The existing emergency bell stays in place and in use. The node does not replace emergency or code systems.
- **Any press alerts the nurse station.** Comfort presses go first to care assistants but stay visible to nurses. A repeated press, or a press from a patient whose current NEWS2 is 5 or more, escalates automatically.
- **Optional fallback beacon.** A direct ESP32-to-ESP32 radio link (ESP-NOW) to a ward beacon so calls still sound when Wi-Fi is down. Range inside the building has to be tested.

4.3 Medication confirmation

- **The nurse confirmation is the administration record.** The nurse taps a badge or scans the wristband and node tag, then confirms "given" in the app. Time and presence are recorded. Checking the right drug and dose stays the nurse's responsibility, and barcode-based verification is a later phase.
- **The patient button is a patient-reported signal.** Pressing "I received my pills" is stored separately, shown as patient-reported, and never closes a dose. It is useful for reminders, for example when a nurse leaves tablets for the patient to take.
- **Time data for the twin.** Presence taps give real time-per-task data for calibrating the simulation. This is opt-in, aggregated and never used to rank individual staff.

4.4 Audio and privacy at the bedside

Pre-recorded phrases in French and Tunisian Arabic, recorded by local staff, give reliable and natural-sounding prompts from a small fixed set. Drug names are never spoken aloud, because rooms may be shared, and details appear on the nurse's device instead. Volume limits and quiet hours protect sleep.

4.5 Reliability and security at the edge

- Store-and-forward queue with unique event IDs, so duplicate deliveries are ignored.
- Real-time clock or server-assigned timestamps, so events logged during an outage keep correct times.
- MQTT with TLS and per-device credentials, on a separate network segment.
- Backup power for the gateway, device inventory and remote updates for nodes.
- Easy-to-clean enclosures suited to a hospital environment.

5. Layers 2 and 3: ward twin and decision modules

5.1 The ward twin

In this document, "digital twin" means a continuously updated model of the ward's state (beds, patients, staff, tasks, equipment) that can be cloned to simulate scenarios without touching live data. Device feeds update the model automatically, and manual entries fill the gaps.

5.2 Decision modules

Module	What it does	Boundaries
Patient priority (NEWS2)	Computes the aggregate early-warning score from fresh, valid vitals and ranks patients on the ward board	A triage aid, not a diagnosis. Observation freshness is always shown. SpO2 scale 2 only on a clinician's decision.
Medication timeline	Builds the schedule from prescriptions, flags overdue doses, detects clashes (for example several infusions due at the same time with one nurse free)	Never selects drugs or doses.
Nurse assignment	OR-Tools balances acuity-weighted workload when something changes	Respects continuity of care, skill mix, proximity, breaks and locked assignments. Always a suggestion for a human to approve.
Equipment reminders	Registry with last-check dates and tickets to the biomedical technician	Software-only reminders at first. Device status from ports comes later.
Explainer	Turns the rule trace behind each flag or suggestion into a plain French or Arabic explanation using templates	No clinical authority. An optional local language model may answer free-text questions.

NEWS2 response thresholds, following the Royal College of Physicians guidance. To be confirmed with a clinical advisor before use:

Score	Meaning	Expected response
0 to 4	Low risk	Routine monitoring
3 in any single parameter	Low to medium risk	Urgent ward-based review
5 to 6	Medium risk	Urgent clinical response
7 or more	High risk	Emergency clinical response

5.3 What-if engine and how it is validated

A discrete-event simulation (SimPy) models patients as entities and nurses as constrained resources. The twin snapshots the ward at a chosen moment, injects a disruption and simulates a few hours forward without altering live data.

- **Example scenarios:** two nurses called away, four urgent admissions at once, an infusion pump failing while a patient deteriorates, a nurse leaving mid-shift.
- **Measurable outputs:** wait time for a nurse, overdue medications, nurse utilization, queue length, longest wait, and patients without observations beyond a threshold.
- **Honest statistics:** each scenario runs over many random seeds and reports the median and the 5th to 95th percentile. Baseline and system run on identical seeds.
- **Honest baseline:** "current practice" is defined from ward interviews, not assumed to be bad.
- **Validation:** first reproduce the queueing-theory result below with the simulator. Then run sensitivity analyses, calibrate with measured timings in a pilot, and compare predictions with observations in shadow mode.

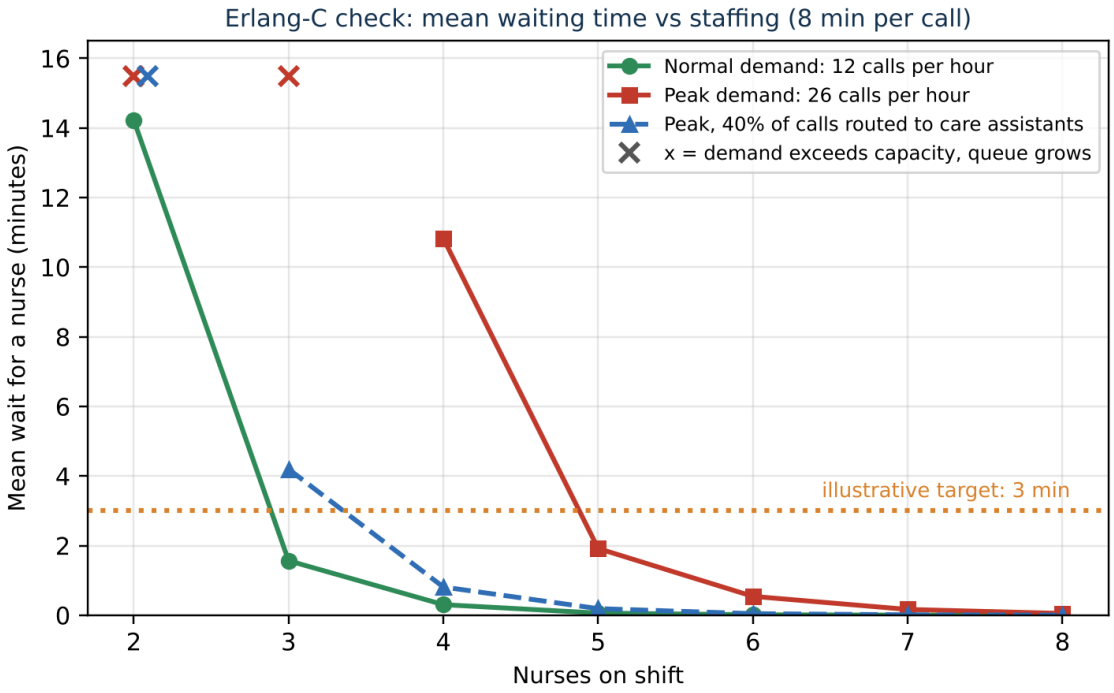


Figure 2. Erlang-C sanity check with illustrative assumptions: 8 minutes per call, 12 calls per hour normally, 26 at peak, and 40% of calls non-clinical in the routed case. This is theory used to test the simulator, not a result for Borj-PiCare.

Illustrative example computed from the same formula (peak demand, mean wait for a nurse):

Nurses on shift	All calls to nurses	40% of calls routed to care assistants
3	unbounded	4.2 min
4	10.8 min	0.8 min
5	1.9 min	0.2 min

The mechanism matters. Reshuffling the same nurses does not change how many people are available. The gain comes from taking non-clinical calls out of the nurses' queue, which assumes care assistants exist on the ward and that the share of non-clinical calls is measured. The classic model assumes steady random arrivals and equal call lengths, which the full simulator will relax. For roster planning, hourly demand profiles from real data are a later addition.

6. Layer 4: staff interfaces

Role	What they get
Nurse	Prioritized task list, medications due, their assignment, calls from their patients
Charge nurse	Ward board, approval of reassignments, what-if panel
Doctor	Escalations at thresholds, prioritized round list, prescription entry
Care assistant	Comfort and service requests, transport tasks
Biomedical technician	Equipment check tickets
Hospital manager (later)	Aggregated workload analytics and what-if studies

The staff app starts as an installable web app and can become native if the team has the skills. Alerts follow an escalation ladder to limit alert fatigue:

- Silent badge for low priority.
- Normal notification for medium priority.
- Persistent alarm that requires acknowledgment for critical events.
- Automatic escalation to the charge nurse or another nurse if nobody acknowledges in time.

Every acknowledgment is logged. SMS is a fallback for anyone without a smartphone, and a large screen at the nurses' station shows the shared ward view. Every suggestion and every decision to accept or override it is recorded.

7. Layer 5 (premium): Family Care Link

7.1 Purpose

Relatives of inpatients want reassurance, and repeated phone calls and visits interrupt care. The Family Care Link gives relatives a controlled window so that staff can answer once, in writing, on their own terms. Whether it really reduces interruptions is a hypothesis. The pilot will count calls to the nursing desk before and after launch.

7.2 What families see and never see

Shown	Never shown
Visiting hours and ward rules Meals served "Medication round completed" without drug names A scheduled, nurse-approved update note Ward contact and emergency instructions	Vital signs, NEWS2 or any risk score Drug names or doses Automatically generated reassurance about the patient's condition Nurse locations or response times Information about other patients

7.3 Consent and access control

- The patient, or a legal representative when the patient cannot decide, chooses who may follow and which items they see. Consent is recorded and can be withdrawn at any time.
- Each authorized relative gets a personal invite by QR code and a one-time code. Invites are bound to the admission, not to the bed, and expire automatically at discharge or transfer.
- Every view is written to an audit log, with rate limits against misuse.

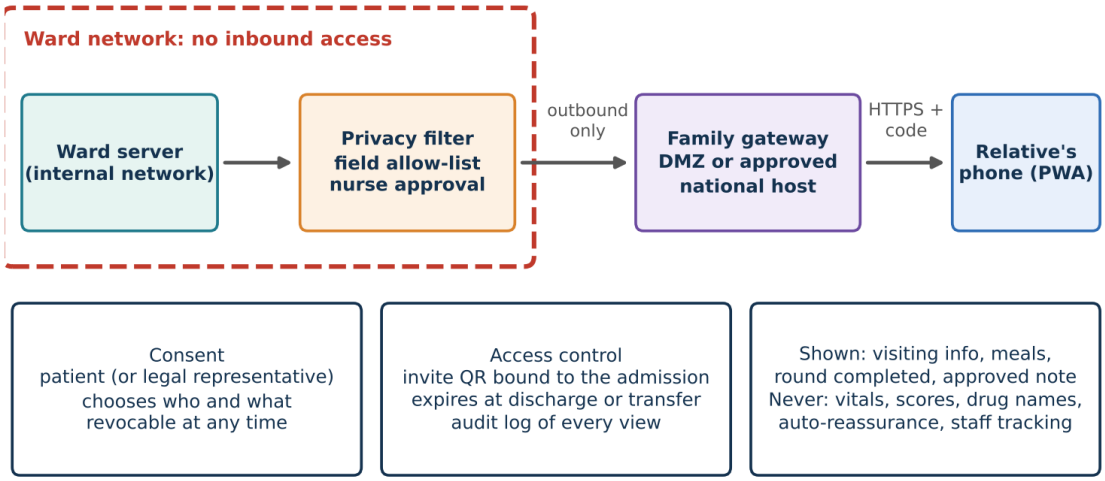


Figure 3. Family Care Link data flow. The ward network publishes filtered updates outward only. Relatives use a standard web app with strong authentication, with no VPN to install.

7.4 Design choices

- **Outbound-only publishing.** A privacy filter applies a field allow-list and nurse approval, then pushes updates to a family gateway in the hospital DMZ or on an approved national host. Nothing on the ward network accepts connections from the internet.
- **No VPN for relatives.** A VPN is impractical for the public, so relatives use HTTPS with strong authentication. A VPN may serve staff maintenance access only.
- **Nurse-approved wording.** Updates are templates or short notes approved by a nurse, and the system never generates reassurance such as "resting comfortably" by itself.
- **Staff protection.** No display of how long ago a nurse last visited, and no staff tracking.
- **Not an emergency channel.** The screen says so and shows the ward phone number.

7.5 Data protection

Even status information such as medication rounds is health-related data about an identifiable person. Under Law 2004-63, health data processing is prohibited by default, with exceptions that require INPDP authorization, informed consent, pseudonymization or anonymization and approved national hosting. The demonstration uses synthetic data only, and any real use waits for authorization and hosting decisions.

7.6 Why it is the premium tier, and who pays

Option	How it works	Strengths	Weaknesses
Facility add-on	The clinic or hospital pays per bed per month and offers the link to patients	Predictable revenue. Clinics can use it to stand out	Needs a paying facility. Public hospitals have little budget
Family-paid	Optional fee per stay, paid by relatives	Revenue grows with use	Sensitive in public hospitals. Payment friction and fairness concerns
Hybrid	Basic updates free, extended features paid (for example scheduled call-back slots)	Fair baseline with an upsell path	More complexity and more moderation

Working hypothesis: the premium tier is most viable first in private clinics, while the core ward tool is pursued for public hospitals through program or grant funding. Prices are set after interviews and pilot data.

8. Risks and mitigations

Risk	Why it matters	Mitigation
Clinical safety	A wrong priority or assignment could harm a patient	Advisory only, human approval, logging, shadow-mode pilot before live use
Device integration varies	Ports exist, but formats, vendor terms and approvals differ by model	Per-model survey, read-only gateway, manual-entry fallback
Health data law	Law 2004-63 prohibits health data processing by default, with authorized exceptions	Synthetic data now. INPDP authorization, consent, anonymization and approved hosting before real data
Regulatory status	Rules for standalone software are unclear in Tunisia	Ask the DPM or ANMPS early. Keep the prototype non-certified and advisory
Adoption	Extra work and alert fatigue sink tools	Co-design, escalation ladder, entry faster than paper, measure and remove what adds work
Lost calls	A call that never arrives is a safety event	Delivery-state feedback, local buzzer, emergency bell stays, optional ward beacon
Family link misuse	Disclosure, false reassurance, conflict with staff	Consent, field allow-list, nurse-approved text, no staff tracking, audit log
Unvalidated twin	Outputs depend on assumed parameters	Queueing-theory check, calibration with measured timings, shadow mode
Procurement and payment	Public buyers can be slow, and the local market is small	Start with private clinics, paid what-if studies and grants
Security	Many connected devices	Network segmentation, TLS, per-device credentials, updates

9. Business model canvas (hypotheses to validate)

Key Partners

- Partner hospital and ward teams (co-design, pilot)
- Biomedical engineers and device distributors (ports, protocols)
- Nursing and medical schools (validation, interns)
- Approved national hosting provider and telecom partners
- Ministry of Health digital teams (identifier, interoperability)
- Funders: grants, startup programs, challenges

Key Activities

- Build and calibrate the ward twin
- Device integration library, model by model
- Shadow-mode pilots and measurement
- Data-protection authorization and security
- Training and support

Key Resources

- Twin and decision engine
- Parsers for device outputs
- Low-cost bedside node design
- Clinical advisors (nurse, physician)
- Authorized, aggregated timing data from pilots

Value Propositions

- **Core, for wards:** prioritized ward board, explainable assignment suggestions, medication timeline, what-if scenarios. Advisory only.
- **For patients:** an easy way to ask for help, with clear feedback.
- **For managers:** evidence for staffing and capacity decisions.
- **Premium, for families:** Family Care Link with consented, nurse-curated updates and less waiting for news (call reduction to be measured).

Customer Relationships

- Co-design with nurses and on-site onboarding
- Support and maintenance plan
- Shared pilot-metric reviews with managers
- Simple invites and help text for families

Channels

- Hospital management and ministry programs
- Healthcare IT forums and startup challenges
- Private clinic groups, directly
- Nursing school and medical society networks
- Word of mouth from the pilot ward

Customer Segments

- Public hospital wards (internal medicine, step-down), reached through hospital management and ministry programs
- Private clinics and clinic groups (first premium buyers)
- Families of inpatients (premium users)
- Later: rehabilitation and long-term care units

Cost Structure

- Development and per-model device integration
- Approved hosting for the Family Care Link
- Hardware kits: nodes, gateway, backup power
- Pilot support, training and travel
- Legal, data-protection and regulatory advice
- Team salaries

Revenue Streams (prices to validate)

- **Core:** per-bed subscription for private clinics, or program and grant funding for public wards
- **Premium:** Family Care Link add-on, paid by the facility per bed per month or by families per stay
- What-if studies for hospital managers (one-off)
- Hardware kits at cost or leased
- Pilot grants and challenge prizes

10. Hypotheses to validate

- Charge nurses and managers find the ward board and the what-if panel useful enough for daily use.
- The ports on the target ward's machines allow read-only capture without objections from vendors or the biomedical team.
- Nurse confirmation taps add little time compared with current paper practice.
- Private clinics will pay for the Family Care Link as a differentiator, and families will use it.
- Calls from relatives to the nursing desk drop measurably after launch.
- Data-protection authorization and approved hosting are obtainable within a pilot timeline.
- Public buyers can fund the core tool through programs or grants, with tolerable payment delays.

11. Roadmap

Phase	Goal
0. Now	Ward interviews, device survey on the target ward, privacy review
1. Hackathon MVP	Twin, NEWS2, assignment suggestions, what-if panel, one bedside node, Family Care Link mockup with synthetic data
2. Shadow pilot	One ward runs the system alongside current practice. Measure timings, calibrate the twin, compare predictions with observations
3. Live advisory use	Advisory use on the ward, and the Family Care Link in a private clinic
4. Scale	More wards and units, integration with national identifier and standards, hospital-level planning

12. Three-minute pitch plan

Time	Content
0:00 to 0:40	The night shift: a charge nurse, a few nurses, many patients, two admissions and a patient worsening at once
0:40 to 1:20	The solution: machine data and bedside nodes feed the ward board, with NEWS2 priority and a medication timeline
1:20 to 2:10	Live what-if: inject a disruption, show the simulator's output for baseline and suggestion, and approve with one tap. Every number on screen comes from the running simulator
2:10 to 2:40	Family Care Link: the consent step, then the limited family view (premium tier)
2:40 to 3:00	Trust and business: advisory only, local data, business model canvas, pilot plan

Questions judges may ask, with honest answers

Question	Answer
Nurses are overwhelmed. Will they use another tool?	Entry is minimized by design: machine data flows in automatically where ports allow, confirmations are single taps, and the fallback form is built to beat paper. We will measure time per task in the pilot and remove anything that adds work.
What if the Wi-Fi fails?	The node shows whether a call was sent. If not, it tells the patient to use the emergency bell. Events are stored and forwarded, an optional beacon sounds locally, and the existing bell stays.
Where does the data come from?	Machine ports where the format is open, verified per model. A short form for fields machines cannot give, such as consciousness. Prescriptions entered by the doctor. Synthetic data in the hackathon.
Is the family link a privacy risk?	It is health-related data, so it is consent-first, limited to an allow-list, nurse-approved, bound to the admission and published outbound-only. Real use needs INPDP authorization and approved hosting. The demo uses synthetic data.
Can the AI make a dangerous mistake?	Scoring and assignment are deterministic. Explanations come from templates built on the rule trace. A language model, if used, only answers free-text questions and has no clinical authority.
Is the twin validated?	Not yet. It compares scenarios. We check the simulator against queueing theory, then calibrate with measured timings and compare predictions with observations in shadow mode.
Does it need regulatory approval?	The rules for standalone software are unclear, so we will ask the DPM or ANMPS. The prototype is non-certified, read-only on devices and advisory.
How is it different from national platforms?	It works at ward level, on live operations. It is designed to connect to the national health identifier and standards later.
Who pays?	Private clinics for the core tool and the premium link, program or grant funding for public wards, and paid what-if studies. Prices are hypotheses to validate.

Appendix A. What changed from version 1

- Hype and unsupported statistics removed. Figures are labeled illustrative, and the typed-in simulation results were replaced by a reproducible analytic example.
- Vital signs now come from a read-only device gateway (USB and Ethernet), with per-model checks and a manual fallback.
- Assist button: delivery-state feedback, fail-visible behavior and a rule that it is never the only route.
- Medication: the nurse's confirmation is the record, and the patient button is a non-authoritative signal.
- Family portal reworked as the premium Family Care Link: consent, allow-list, nurse approval, admission-bound access, outbound-only publishing, and no VPN for relatives.
- NEWS2 thresholds completed. ESP32 nodes with a Pi gateway replace a Pi per bed. Pre-recorded audio replaces synthetic voices. The language model is optional. The FFT module and burnout index were dropped. Business model canvas, hypotheses, risks, roadmap and honest answers added.

References

INPDP / Council of Europe health data guide (Law 2004-63): rm.coe.int/sensibilisation-a-la-protection-des-donnees-personnelles-dans-le-secte/1680a8dd45

AFD e-health program: afd.fr/fr/projets/reequilibrer-lacces-aux-soins-en-deployant-une-strategie-e-sante-dans-les-hopitaux-publics

Sahetna.tn and the national health identifier: fr.allafrica.com/stories/202607210663.html

How digital twins can improve health system operations (JMIR, 2026): jmir.org/2026/1/e98113

NEWS and NEWS2 outcomes review: arcwestarchive.org.uk/news/evidence-lacking-on-effectiveness-of-national-early-warning-scores

Startup funding and public-sector payment challenges: menastartupdigest.com/tunisias-startup-act-hits-1352-certified-companies-but-funding-and-red-tape-challenges-remain

Medical device registration overview: meddeviceguide.com/blog/tunisia-dpm-medical-device-registry-analysis